Statistics – Scenario-Based Questions

Part 2: Analysis approach for real-world data problems

Scenario 1: Flight Delay Analysis

Question: An airline tracks flight delays (in minutes) for 20 flights. How do you analyse the flight delays to calculate percentiles, detect outliers and evaluate the overall distribution?

Answer:

- **Percentiles:** Calculate the 10th, 25th, 50th (median), 75th and 90th percentiles to see how the delays are spread across different levels.
- **IQR:** Compute IQR = Q3 − Q1. A value is treated as an outlier if it lies below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR.
- **Distribution:** A **box plot** shows the spread, median, quartiles and any outliers. A **histogram** shows the shape of the distribution, its skewness and how often each range of delay occurs.

Scenario 2: Employee Salary Analysis

Question: A company wants to analyse the salary distribution of its employees to understand the central tendency and check whether the data is skewed. How should this be done?

Answer:

- **Central tendency:** Calculate the mean, median and mode of the salaries.
- **Skewness check:**
  - Mean > Median → **right-skewed** (a few high salaries pull the mean up).
  - Mean < Median → **left-skewed** (a few low salaries pull the mean down).
  - Mean ≈ Median → **symmetrical**.
- **Best representation:** Use the **median** when outliers are present, because it is not affected by extreme values.

Scenario 3: Product Sales Analysis

Question: A retail store records product sales over 15 days. How do you create a frequency distribution table and visualise the sales data?

Answer:

- **Frequency distribution:** Divide the sales figures into equal intervals (for example 5 or 10 units) and count how many days fall in each interval.
- **Visualisation:**
  - **Histogram** – shows the frequency of sales in each interval.
  - **Bar plot / line plot** – shows day-by-day sales so changes over time are easy to spot.

Scenario 4: Student Exam Performance Analysis

Question: A school wants to analyse the exam performance of students in three subjects – Mathematics, Science and English. How can data science concepts be applied to understand their performance?

Answer:

1. Data preprocessing

- Handle missing values by filling them with the mean or median.
- Convert categorical values (if any) using label encoding or one-hot encoding.

2. Descriptive statistics

- Calculate the mean, median, mode and standard deviation for each subject to understand central tendency and spread.

3. Visualisation

- Box plots to detect outliers.
- Histograms to see the distribution of scores.
- Scatter plots to check the relationship between subjects.

4. Correlation analysis

- Compute correlation coefficients to see how the subject scores relate to each other.
- Use a heatmap to visualise the correlation matrix.

Scenario 5: Clinical Trial for Diabetes Medication

Question: A pharmaceutical company ran a clinical trial with two groups – one received the medication and the other a placebo. How do you perform a hypothesis test to find out whether the medication is effective?

Answer:

- **Set up the hypotheses:**
  - **H_0 (null):** There is no difference between the medication and the placebo.
  - **H_1 (alternative):** The medication lowers blood sugar more than the placebo.
- **Run an independent two-sample t-test** and compare the p-value with 0.05:
  - $p \leq 0.05 \rightarrow$ reject H_0; the medication is effective.
  - $p > 0.05 \rightarrow$ fail to reject H_0; no significant difference.